Local rule activation and dealer capacity in a cellular automaton market*

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Abstract

Changes in investor decisions can spread through a market, but the resulting price decline depends on how trades execute. We developed a market model in which traders inside a grid decided whether to buy, hold, or sell from their neighbors' decisions, while technical traders at selected perimeter sites responded to price. Paired markets shared the same history before selected traders in one market received a persistent change in their local decision rule. The comparison market retained its original rule. A majority decided to sell when enough traders received the new response, even though the remaining traders kept their original rules. Holding the technical traders prevented price changes from entering later decisions, yet majorities still formed, establishing that neighbor responses could produce widespread selling. Comparisons across market sizes and observation periods located a finite-time crossover whose position continued to depend on the time observed. To connect decisions to prices, we cleared unit orders against a dealer whose finite cash and share reserves retained a constant product. The batch execution price and resulting marginal quote were both determined by those reserves, preventing an isolated round trip that restored dealer inventory from extracting its cash. Shallow positive reserves accompanied severe declines while rejecting few orders backed by trader resources. Zero reserves instead permitted only opposing trader orders at a fixed price. Shallow markets also had large drawdowns without the rule switch, making matched financial comparisons necessary for attribution. Changing trader cash and shares affected interior decisions only when technical traders could respond to the resulting prices. The model separated the spread of selling decisions from the severity of a simulated crash. Limited trading resources did not generally protect against severe declines, because their financial consequences depended on the mechanism linking execution to price.

*Repository: https://github.com/varnerlab/Wolfram-markets-abm-paper.git.

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1 Introduction

Financial markets are the collective outcome of human interactions. They reflect both our ingenuity and our failings. In its 2011 report on the 2008 financial crisis, the Financial Crisis Inquiry Commission paraphrased Shakespeare: "the fault lies not in the stars, but in us" [1, p. xvii]. In markets for traded financial assets such as stocks, investors make decisions with different information, expectations, and financial resources. They respond to prices and to the actions of others when deciding whether to buy, hold, or sell. An investor's decision to sell can prompt others to do the same. Sellers need buyers who are willing and able to purchase their assets. The prices buyers will pay reflect their views of the assets' value and the funds they can commit to the purchase. If sellers cannot find buyers at the current price, they may accept lower prices to complete their sales. The resulting price changes can then influence investors' later decisions. Under what conditions can a change in the behavior of some investors lead to widespread selling, and when does that selling produce a market crash?

Agent-based models study how individual trading decisions and interactions among investors produce market behavior. This literature includes many approaches to learning, imitation, and price formation [2, 3]. Models differ in the information available to traders and in how their trades affect prices. Farmer and Joshi studied traders who followed price trends or compared prices with their estimates of asset value. A market maker absorbed their net orders and adjusted the price. The resulting price changes then influenced later trading decisions [4]. In the Santa Fe artificial stock market, traders learned forecasting rules from market prices and dividends and used those forecasts to guide their trades [5]. In previous work, we developed an agent-based market that supported trading in multiple assets. Agents submitted orders to a continuous double auction, which matched compatible buy and sell orders as they arrived. Traders with different decision rules therefore interacted through a common trading mechanism [6]. Alongside interactions through prices and trades, herding models describe how traders respond to the behavior of others [7–9]. Arranging traders in rows and columns provides a simple representation of the connections through which influence operates. Each position represents a trader, and neighboring positions identify the traders whose decisions it can observe. The information passed through these connections can be as simple as a trader's current decision to buy or sell [10]. A change in one trader's decision can affect its neighbors, whose responses can then influence other traders. Models inspired by the Ising model use these local interactions to represent imitation. In Bornholdt's simplified model, traders tended to follow their neighbors. A growing imbalance between buyers and sellers across the market increased their tendency to reverse their current decisions [11]. Iori used a related model in which traders could buy, sell, or remain inactive. Traders repeatedly exchanged their current decisions with their neighbors and used the resulting signals to revise their own choices [10]. These models used similar arrangements of traders but different rules for responding to neighboring decisions. The arrangement specifies who can influence whom, while the decision rule specifies how each trader responds.

Cellular automata use local rules to update traders' decisions at discrete time steps [12, 13]. Wolfram arranged traders on a line. Each trader responded to its two neighbors, and the cumulative difference between buying and selling served as a price analogue [14]. Later models combined cellular automata with trader balance sheets [15] or used stochastic rules for interactions between neighboring traders [16–20]. Our model represents a synthetic market with traders arranged in rows and columns. Two types of traders occupy the market interior. Both types use their four neighbors' intended actions to decide whether they want to buy, hold, or sell. Their rules differ in a single response to their neighbors' intended actions. We vary the relative numbers of the two types to study how the mixture affects the spread of selling intentions. Technical traders along the edge of the synthetic market base their decisions on prices. Their choices provide a path through which price changes can influence interior traders. The financial consequences of intended trades depend on the resources available for exchanging cash and shares. Earlier agent-based models have examined limits on traders' cash and asset holdings and on their ability to borrow [21, 22]. In our model, Buy and Sell indicate intended actions. A completed trade transfers shares from a seller to a buyer and cash from the buyer to the seller. Traders cannot borrow cash or sell shares they do not own. A trader without shares can still indicate a desire to sell but cannot carry out the sale. A trader without enough cash can indicate a desire to buy but cannot carry out the purchase. Neighboring traders respond to the intended actions even when no trade occurs. Selling intentions can therefore spread without a corresponding increase in completed sales. A dealer acts as the market maker. It can use its finite cash and share reserves to complete trades that cannot be matched between traders. Completed trades preserve the product of the

dealer's cash and share reserves, following a constant-product market-making approach [23]. After each round of trading, the dealer's quoted price per share equals its remaining cash divided by the number of shares it holds. When the dealer buys shares from traders, it has less cash and more shares. Dividing the smaller cash balance by the larger number of shares gives a lower quoted price. The size of the dealer's reserves determines both which trades can be completed and how much the quoted price changes when the dealer buys or sells shares. Technical traders use the dealer's updated quote to make their next decisions. The technical traders' decisions then influence neighboring interior traders.

In this study, we constructed a synthetic market based on a three-state cellular automaton to examine whether mixing two trader types could lead to widespread selling intentions. We also examined how trader holdings and dealer reserves affected the accompanying price decline. In each paired rule-change comparison, both markets had the same number of traders and shared a period of trading with only the original trader type in the interior. We then assigned the second decision rule to a chosen fraction of interior traders in one market. The other market kept the original rule. Trading intentions, trader holdings, dealer reserves, and prices matched when the rule changed. The traders could then make different decisions and complete different trades. We varied the fraction of interior traders using the second rule to determine when at least half of the interior traders wanted to sell, which we called a Sell majority. In separate experiments, we varied the total number of traders and the simulation duration to examine how market size and time affected the spread of selling. We also ran markets with all technical traders held in Hold to test whether a Sell majority could develop without feedback from prices. Interior traders continued to respond to neighboring intentions and could complete trades with one another or the dealer. Trades with the dealer could still change its reserves and quoted price. In the rule-change experiments, keeping technical traders in Hold prevented price changes from influencing interior traders' intentions. However, prices still affected which trades could be completed. Buyers needed enough cash to purchase shares, and sellers needed shares to sell. Each sale also required a buyer or dealer able to pay the execution price. We varied dealer reserves and traders' cash and shares separately to examine how those resources affected completed trades and price declines. We also examined crowding by having groups of traders follow shared advice or copy another trader's completed trades. We varied the fraction of followers and the number of sources to examine effects on other traders' decisions, prices, and the distribution of gains and losses. For shared advice, we tested whether followers' influence on neighboring traders contributed to their financial gains. Together, the rule-change and following experiments showed how individual trading decisions could spread through a market, affect prices, and redistribute cash among traders.

2 Methods

2.1 Three-state cellular automaton

We represented the synthetic market as L rows and L columns, with $(L-2)^2$ traders in the interior. We recorded each trader's intended action as Buy ($s = 0$), Hold ($s = 1$), or Sell ($s = 2$). Technical traders occupied selected positions along the edge and formed their intentions from the dealer's quoted price and recent price history. The remaining edge positions stayed in Hold. Each interior trader formed its local decision from the intentions of its four neighbors above, below, to the left, and to the right. The decision rules were totalistic: each trader's next intention depended only on the sum of these four neighboring states [13]. For interior traders next to the edge, this sum could therefore include the intentions of technical traders. At each time step t, the technical traders updated their intended actions using the latest quote and price history. All interior traders then updated their intended actions together using the previous interior states and the newly updated edge states. We denoted the four neighbors of trader i by ∂i and the state supplied by neighbor j at step t by $s_j^{\text{in}}(t)$. The totalistic rule used their sum $c_i(t)$, defined by:

$$c_i(t) = \sum_{j \in \partial i} s_j^{\text{in}}(t) \in \{0, 1, \ldots, 8\}. \tag{1}$$

We specified each decision rule by nine outputs a_k, one for each neighbor sum $k \in \{0, 1, \ldots, 8\}$. Each output recorded the intended action assigned to that sum. The Wolfram rule number R encoded these outputs as base-three digits [12, 13]:

$$R = \sum_{k=0}^{8} a_k 3^k. \tag{2}$$

Rule 19280, $(222110002)_3$, supplied the baseline decision rule. The activated Rule 19286, $(222110022)_3$, differed only at $c = 1$ (Eqs. 1 and 2). A neighborhood containing three Buys and one Hold therefore returned Buy under the baseline and Sell under activation. Before each run, we selected $\mathrm{round}(f(L-2)^2)$ interior sites uniformly without replacement as the target set for the later switch. The reported fraction f was nominal; its realized value followed this integer rounding. The initial interior contained $\mathrm{round}(0.05(L-2)^2)$ Buy sites and the same number of Sell sites, with Hold elsewhere. Initial states were placed without replacement. A random-number seed identified each starting market and fixed the initial placements, selected traders, technical traders, and their band parameters. It also fixed the priorities used when eligible orders exceeded dealer capacity. Once these choices were made, the market evolved deterministically. Delaying activation allowed the market to develop under the baseline rule before the response changed. At the intervention, each targeted trader retained its current decision and holdings but used the new rule to make subsequent decisions. Traders outside the target set kept their original decision rule throughout.

2.2 Orders, trader resources, and completed trades

A trader's decision generated a desired order, but completing it required cash or shares and a counterparty. Every interior trader and technical trader held cash w_i, an integer number of shares q_i, and average purchase price $\bar{c}_i$. Buy requested one share, Sell offered one share, and Hold requested no trade. All completed orders in a step used one execution price, determined together with the quantities that could trade. A buy was eligible only if the trader could pay that execution price, and a sell required at least one share. The main experiments disabled the optional restriction on selling below average acquisition cost [24, 25]. We recorded desired orders lacking trader resources as *agent suppression*. A trader's stored decision persisted even if its order failed: a trader could remain in Sell without shares, and its neighbors still responded to that state. Thus financial resources constrained execution without directly changing the interior decision rule.

The dealer supplied unmatched orders from finite cash and share reserves. Capacity h specified initial dealer shares per trader. For N_{tr} traders and initial price $P_0 = 100$, the dealer started with $H = \mathrm{round}(hN_{\mathrm{tr}})$ shares and cash $P_0 H$. For positive h, its cash W and shares Q followed a constant-product reserve curve, a two-asset instance of constant-function market making [23]. The dealer could neither borrow cash nor sell borrowed shares, and retained at least one share. For a candidate net purchase x by traders, positive for purchases from the dealer and negative for sales to it, the uniform execution price $F_t(x)$ was the cash transfer per net share needed to preserve their product K:

$$K = W_{t-1}Q_{t-1}, \qquad Q_{t-1} - x \geq 1,$$
$$F_t(x) = \frac{W_{t-1}}{Q_{t-1} - x}. \tag{3}$$

At this candidate price, $B_t(x)$ counted desired buys backed by enough cash and $S_t(x)$ counted desired sells backed by shares. Their difference at $x = 0$ determined direction d_t. For nonzero direction we selected the largest supported integer quantity, denoting the eligible counts at the selected price by B_t and S_t. The selected net purchase was x_t, and n_t^B and n_t^S denoted executed trader buys and sells:

$$d_t = \mathrm{sign}\{B_t(0) - S_t(0)\},$$
$$x_t = d_t \max\left\{k \in \mathbb{Z}_{\geq 0} : Q_{t-1} - d_t k \geq 1, \quad d_t[B_t(d_t k) - S_t(d_t k)] \geq k\right\}, \tag{4}$$
$$n_t^B = \min\{B_t, S_t + x_t\}, \qquad n_t^S = n_t^B - x_t. \tag{5}$$

For $d_t = 0$, we set $x_t = 0$. The smaller eligible side completed every order, and reproducible priorities selected orders on the larger side when it could not be filled completely. We recorded eligible orders left unfilled as *dealer rejection*. Rejection could reflect the final share constraint or the inability to support another indivisible share at its required price; it did not require zero dealer cash.

All executed trader orders settled at $F_t = F_t(x_t)$, including trades matched between traders. Settlement changed the dealer's books and its next marginal quote P_t according to:

$$Q_t = Q_{t-1} - x_t, \qquad W_t = W_{t-1} + F_t x_t, \tag{6}$$

$$P_t = \frac{W_t}{Q_t} = P_{t-1} \left(\frac{Q_{t-1}}{Q_{t-1} - x_t} \right)^2. \tag{7}$$

The execution price and the resulting marginal quote therefore differed when net dealer trading was nonzero. Cash and shares were conserved across traders and dealer, except at scheduled changes in trader resources. In exact arithmetic, $W_t Q_t = K$ made an isolated sequence returning dealer inventory to its initial value return dealer cash as well. Such a round trip could not extract cash from the dealer. This identity concerns isolated net trades; it does not rule out gains from other traders. We applied no additional exponential price adjustment. Reserve depth consequently determined both funding and sensitivity of price to net trades. The deep comparison used finite $h = 100$ throughout. At $h = 0$, opposing trader orders matched at the unchanged quote, with no net dealer trade. This separate case supplied a fixed-price control (Algorithm S2).

2.3 Technical traders and price feedback

Selected traders along the edge of the grid brought price information into neighboring decisions. They followed a rule of the Bollinger-band type: each bought below a band around a recent price average, sold above it, and held inside. This rule was contrarian, in contrast to the trend-following moving-average rules often tested in market data [26]. One quarter of perimeter cells held technical traders; the other perimeter cells stayed in Hold. A technical trader compared the new marginal quote with reference price μ_t and a band whose width used recent absolute price departures σ_t. Each trader retained a multiplier $k_j \sim U[0.5, 1.5]$, giving lower and upper limits $\mu_t - k_j \sigma_t$ and $\mu_t + k_j \sigma_t$. The normalized net purchase ϕ_t described completed orders but did not separately set price. With smoothing parameter $\alpha = 1/60$, the recorded imbalance and updated band state were defined by:

$$\phi_t = \frac{x_t}{N_{\mathrm{tr}}}, \tag{8}$$

$$\mu_t = \mu_{t-1} + \alpha(P_t - \mu_{t-1}), \qquad \sigma_t = \sigma_{t-1} + \alpha(|P_t - \mu_t| - \sigma_{t-1}). \tag{9}$$

Orders came from the stored decisions at the start of each step. Scheduled rule and resource changes preceded order collection, clearing, and settlement. The new reserve ratio then set the marginal quote, technical traders responded to it, and interior traders updated together using old interior decisions and the new technical-trader decisions (Algorithm S1). The complete step followed the sequence:

stored decisions $\to$ orders $\to$ batch clearing $\to P_t \to$ technical decisions $\to$ interior update.

We prevented price feedback by setting every technical trader to Hold before each interior update. We call this condition *technical traders held*; *technical traders active* denotes their response to the new quote. Figure legends abbreviate these conditions as held and active. The comparison retained the same initial grid, target set, technical-trader locations and parameters, and financial resources. Every held market still used the complete financial simulator. Cash, shares, and price could change, but those changes could no longer influence decisions inside the grid. Comparing held and active markets therefore tested whether a Sell majority could develop from responses to neighbors without feedback from prices.

2.4 Paired rule-switch experiments

We compared the consequences of changing a decision rule with continued use of the original rule in paired markets that shared a common history. Both paths used Rule 19280 at every interior site for $T_{\mathrm{pre}} = 1{,}000$ baseline steps. At the start of step $T_{\mathrm{pre}} + 1$, one path activated Rule 19286 at the preselected fraction f. The matched control left those sites on Rule 19280. Neither path reset the lattice, price, technical traders' bands, trader holdings, or dealer. Their common state $X_{T_{\mathrm{pre}}}$ also generated the first price after activation, so the earliest possible difference occurred in the following interior update. In the main paired rule-switch campaigns, we checked the shared pre-switch histories and first post-switch price bit for bit. Each switched path thus began with the same trader resources and neighbor configurations as its control. The four main sets of simulations used 50 seeds and

3,000 steps after the rule change, with initial trader cash of 1,000 and ten shares each. Additional comparisons tested other rules, a one-time instruction to sell, fixed widths for the technical traders' bands, additional reserve depths, and longer observation periods. The supplementary protocols specify which part of the intervention or feedback was changed in each control (Appendix A.5).

We varied the fraction of traders given the new rule, grid size, and financial resources to determine when a Sell majority formed and how many sales could complete (Supplementary Table S1). The selected switch experiment used $L = 64$, a dealer with $h = 100$, and active technical traders at four activation fractions. We then tested more fractions at eight dealer capacities under both technical-trader conditions. The finite-size campaign used $L \in \{32, 48, 64, 96, 128\}$ with the technical traders held and $h = 100$, or active and $h \in \{1, 100\}$. Because capacity was defined per trader, absolute dealer resources increased with lattice size. The financial campaign multiplied current trader cash and shares independently by $g_w, g_q \in \{0.1, 1, 10\}$ at activation, rounding shares to integers. Its nine interventions shared the same pre-intervention path within each $(f, h, \text{technical traders}, \text{seed})$ cell. With the technical traders held, they also had to retain identical lattice trajectories afterward because neither cash nor shares entered the interior update directly. A discrepancy would therefore indicate an implementation error. The appendix gives the full parameter grids and the checks used to verify these paired relationships and the identities required when the technical traders were held.

2.5 Market outcomes, statistical analysis, and selection of examples

We used the strictly interior Sell fraction $m(t)$ to measure the shift toward selling. Larger values indicated that more interior traders occupied Sell, and we treated the emergence of a Sell majority after the rule change as a sign of market instability. We recorded a Sell majority when $m(t) \geq 0.5$ at least once after the rule change. The onset time was the first step that reached this value, and majority probability was the fraction of runs that reached it. We summarized onset times only for runs that reached a majority and averaged the Sell fraction over the final 500 steps to describe each market near the end of the observation period. To measure how much this final mean $\bar{m}$ differed between runs, we multiplied its variance by the number of interior traders, $(L-2)^2$, with angle brackets denoting averages across seeds. This scaled variance is the susceptibility χ_L, which allows comparison across grid sizes [27, 28]:

$$\chi_L = (L-2)^2 \left[\langle \bar{m}^2 \rangle - \langle \bar{m} \rangle^2\right]. \tag{10}$$

At a fixed grid size, higher susceptibility indicated greater variation in the final mean Sell fraction between starting markets (Eq. 10). Linear interpolation gave $f_q(L)$, the activation fraction at which the transition probability reached $q/100$. The midpoint was $f_{50}(L)$, and the width $f_{75}(L) - f_{25}(L)$ measured how much additional activation separated probabilities of 0.25 and 0.75. A smaller width therefore indicated a sharper response. Interpolation used the first upward crossing without forcing the sampled curve to be monotone. Binomial proportions used 95% Wilson intervals [29]; continuous summaries used interquartile ranges unless otherwise specified.

A high Sell fraction need not produce a large price decline if the requested orders cannot execute. We measured the decline by maximum drawdown, the largest proportional fall from a running price peak during the post-activation window. We used price crash to describe a large simulated decline and reported its magnitude through drawdown, without defining a cutoff for classifying runs as crashes. We recorded unfilled orders at the stage where resources prevented execution. Agent suppression counted desired orders excluded by insufficient trader cash or shares. Dealer rejection counted eligible orders left unfilled at the selected execution price. In every campaign, the rejection fraction divided post-activation rejected orders by post-activation eligible orders. Agent suppression and completed-order counts used the same window. The price window included the quote immediately before the rule change, while majority detection began with the first updated interior state after that change. Onset therefore began at step one; a missing onset indicated no crossing within the observation period. Financial summaries used medians and interquartile ranges across seeds. Drawdown measured the price decline caused by completed trades, while suppression and rejection identified why desired trades could not complete.

Testing too few fractions of traders with the new rule could miss the range over which majority probability changed. We refined each curve using a common resolution criterion. Intervals intersecting the empirical 5–95% response were bisected for up to ten rounds, aiming for at least five fractions strictly inside that probability range in each size and condition. Unbracketed crossings and unresolved curves remained explicitly missing or flagged; observed outcomes were not used as pass conditions.

Each added fraction used the original 50 seeds. We then resampled whole seeds across all fractions 2,000 times to estimate uncertainty in the crossings, width, and sampled susceptibility maximum. These percentile intervals preserved dependence between fractions and included reselection of the susceptibility peak. We also fitted a logistic curve, which required majority probability $p_L(f)$ to increase smoothly with the fraction f. The fitted parameters a_L and b_L controlled its position and slope:

$$p_L(f) = \{1 + \exp[-a_L - b_L(f - 0.18)/0.01]\}^{-1}, \qquad b_L > 0.$$

We fitted this curve by binomial likelihood and used the same seed blocks to bootstrap its midpoint and width. Fitted standard errors were bootstrap standard deviations. Both analyses measured uncertainty for the tested fractions and observation period. Neither included error from interpolation, the choice of additional fractions, or observing the market for longer.

We selected examples of traders' decisions across the grid using the measured time to a majority or final Sell fraction to choose each run. Selection preceded inspection of the images, with the smaller seed breaking equal distances from a group median. The newly computed 3,000-step results supplied a run without a majority at $f = 0.15$, both outcomes at $f = 0.175$, and runs with a majority at $f = 0.185$ and 0.20 under $h = 100$ (Fig. 4). Among runs that reached a majority, we selected the run nearest the group's median onset time. Among runs without a majority, we selected the run nearest the group's median final Sell fraction. The dealer panels required a single seed that transitioned at $h = 0.1$, 1, and 100 for $f = 0.20$. Among eligible seeds, selection minimized the summed distance from the three capacity-specific median onsets. We extended the five examples across fractions assigned the new rule to 10,000 steps after the change, with snapshots at 0, 1,500, 3,000, 5,000, and 10,000 steps (Fig. 4). The dealer comparisons retained their original 3,000-step duration and snapshots at 0, 500, 1,500, and 3,000 steps. Selection therefore represented typical outcomes within each group over the original observation period. A market without a majority at that endpoint could reach one later. Individual runs illustrated how groups of sellers developed, while probabilities across runs measured how often a majority occurred.

3 Results

To explore how a persistent change in investor decisions could destabilize a market, we changed one response in selected traders' rules and found that a Sell majority formed when enough traders received the new rule (Fig. 1; Supplementary Tables S1 and S2). Rule 19286 changed the response to three neighbors in Buy and one in Hold from Buy to Sell, preserving every other response (Fig. 1A,B). Each treated market shared its baseline history and first post-switch quote with a comparison market that retained Rule 19280. With active technical traders and dealer depth $h = 100$, none of the 50 starting markets reached a majority at $f = 0.10$ or 0.15, whereas all reached one at $f = 0.20$ and 0.25. No matched control reached a majority over the same period. The median onset among the two larger fractions changed from 1,070.0 to 311.5 steps (Supplementary Table S2). In the selected pair, groups of sellers expanded after the rule change while the comparison market retained mixed decisions (Fig. 1C). The lasting rule change also differed from a temporary instruction: setting the selected traders to Sell once at $f = 0.15$, 0.20, or 0.25 produced no majority in either technical-trader condition (Supplementary Table S5). Among the eighteen single-response rule changes tested at $f = 0.20$, 6 produced a majority in every tested starting market under both feedback conditions, while 12 produced none (Supplementary Table S6). Within these tested interventions, majority formation differed with the changed response, the fraction of traders receiving it, and whether the intervention changed a rule or only the current decision.

To determine whether price feedback was required for widespread selling, we repeated the rule switch while keeping technical traders in Hold and found that a majority still formed (Fig. 2). With these traders held, majority probability was 0.34 at $f = 0.175$, 0.70 at $f = 0.185$, and 1.00 at $f = 0.20$ (Fig. 2A). Every held market was simulated again with its full cash, share, and price history. Across dealer depths, onset, final mean Sell fraction, and final interior states were identical for the same starting market and target fraction, although the associated financial paths differed. This agreed with the held update, which had no input from financial variables. Neighbor responses were therefore sufficient to produce a Sell majority. Allowing technical traders to respond to price changed majority frequency near the crossover between mostly mixed and mostly Sell-majority outcomes. At $h = 100$, the active condition gave 25 of 50 majorities at $f = 0.175$ and 42 at $f = 0.185$, compared with 17 and 35 when held (Fig. 2A,B). Both the size and direction of the active-minus-held difference depended

on the tested depth and fraction. A price response could thus modify how often the rule change produced a majority, but it was not necessary for selling decisions to spread. These comparisons separated the local decision mechanism from feedback through completed trades.

To determine how the spread of selling depended on the number of interacting traders, we compared grid sizes with technical traders held and found a sharper majority response on larger grids (Fig. 3). We refined the fractions near each response while retaining the same starting markets. At the original 3,000-step horizon, the interpolated fraction at which half the markets reached a majority, f_{50}, changed from 0.216 at $L = 32$ to 0.166 at $L = 128$. The width separating majority probabilities of 0.25 and 0.75 changed from 0.023 [0.018, 0.027] to 0.008 [0.004, 0.010], with brackets giving 95% intervals from resampling whole starting markets (Fig. 3A,C; Supplementary Table S3). Variation in the final mean Sell fraction also changed: its variance multiplied by the number of interior traders, the susceptibility, had sampled maxima of 21.2 and 241.2 at the endpoint sizes (Fig. 3B,D). Across all size and feedback conditions, the largest absolute difference between logistic and interpolated midpoints was 0.0035, indicating the sensitivity of the estimate to the imposed curve shape (Supplementary Tables S3 and S10). Observation time remained consequential. At $L = 64$, extending the held markets from 3,000 to 9,000 steps changed the majority midpoint from 0.178 to 0.167; alternative Sell thresholds and size-dependent horizons were measured separately (Supplementary Table S9). The size response therefore described a crossover within the observation window, while the continued movement with time prevented its interpretation as a fixed threshold for indefinitely long runs.

To examine how the spatial patterns developed beyond the initial observation window, we extended five numerically selected markets and found that longer observation changed some outcomes (Fig. 4). Selection used the newly computed 3,000-step results and preceded inspection of images. Two selected markets had not reached a majority at that endpoint, while three had. At 10,000 steps, 1 of the initially nonmajority examples had crossed one-half. The first-majority outcomes, in displayed order, were $f = 0.150$, starting market 2: not within 10,000 steps; $f = 0.175$, starting market 15: after 4,073 steps; $f = 0.175$, starting market 41: after 2,330 steps; $f = 0.185$, starting market 5: after 1,836 steps; $f = 0.200$, starting market 38: after 1,028 steps (Supplementary Table S11). The spatial histories connected these crossings to the growth and merger of groups of sellers, while the price histories measured their financial consequences (Fig. 4A–C). In the selected $f = 0.175$ market that had not crossed by 3,000 steps, maximum drawdown changed from 10.17% to 26.34% at the longer endpoint (Supplementary Table S11). We also extended all 50 starting markets at $f = 0.150$, retaining matched no-switch comparisons. The number of majorities among changed-rule markets was 0 at 3,000 steps and 2 at 10,000 steps, while 0 controls reached a majority by 10,000 steps. Median drawdown changed from 5.04% to 5.87%, and the largest observed drawdown changed from 12.29% to 25.30% (Supplementary Table S11). The longer trajectories therefore qualified outcomes defined by the original endpoint; persistence of mixed decisions over that shorter window did not establish persistence over longer times.

We varied dealer depth while retaining the same decision-rule intervention and found that shallower positive reserves accompanied larger price declines, while majority probability did not vary monotonically with depth (Fig. 2). At $f = 0.20$, every starting market reached a majority at every tested depth. With active technical traders, median maximum drawdown was 99.99% at $h = 0.1$, 99.52% at $h = 1$, and 25.15% at $h = 100$ (Fig. 2C). The corresponding median rejection fractions were 0.000226, 0.000034, and 0.0000006, so small positive reserves supported large declines while rejecting few orders backed by trader resources (Fig. 2D). Zero depth supplied a separate limit: only opposing trader orders matched, net dealer trading was zero, and price stayed fixed. No-switch markets at $h = 0.1$ already had median drawdown 77.04%, compared with 0.75% at $h = 100$. The median paired increase after the rule switch was 22.96 percentage points at $h = 0.1$ and 24.39 at $h = 100$ (Supplementary Table S2). Positive reserve depth changed both available funding and the price required to execute a net trade. In the additional comparison at $h = 40$ and $h = 400$, median drawdowns were 48.62% and 7.08%, respectively (Supplementary Table S8). Near the majority crossover, depth also changed feedback through the technical traders. At $f = 0.175$, majority probabilities were 0.44 at $h = 0.1$ and 0.50 at $h = 100$. Their difference changed from -0.06 with adaptive band widths to -0.26 with frozen widths (Supplementary Table S7). The paired difference between those contrasts was 0.20, with a 95% bootstrap interval [-0.08, 0.50], so its sign was not resolved (Supplementary Table S7). The selected comparison from starting market 2 also reached a majority at all three displayed depths while following distinct price paths (Fig. 5). The reserve mechanism

thus separated majority formation from the severity of the decline, but shallow positive reserves did not prevent severe drawdowns.

To distinguish trader resources from dealer reserves, we changed cash and share holdings independently at the rule switch and found that financial resources changed interior decisions only through active price feedback (Fig. 6). At $f = 0.18$, all cash and share combinations with technical traders held produced the same decision history within a starting market, giving majority probability 0.64 across dealer depths. With technical traders active, probabilities ranged from 0.54 to 0.80 (Fig. 6A; Supplementary Table S4). Holdings entered the order and clearing calculations but did not enter the interior lookup table. Their effect on subsequent decisions therefore required the intervening price response. With unchanged cash and $h = 100$, share multipliers 0.1, 1, and 10 gave median drawdowns of 3.17%, 20.18%, and 77.55%, respectively (Fig. 6B). More shares supplied more potential sales, while the reserve curve determined the price at which net selling could clear. Increasing the cash multiplier from 1 to 10 at $h = 1$ with unchanged shares changed median drawdown from 96.46% to 97.10%, and majority probability from 0.58 to 0.56 (Fig. 6A,B). Rejection recorded the fraction of orders backed by trader resources that still failed to execute, and its relationship to drawdown differed across the resource combinations (Fig. 6C,D). The financial consequences of a rule change therefore depended on both sides of traders' books and the dealer's reserves. A majority decision to sell alone did not specify how many shares could change hands or how far the marginal quote would fall.

4 Discussion

Changes in investor decision rules spread selling through the market, while trader resources and the dealer's reserve curve determined the associated price decline. A majority entered Sell after enough traders received a new response to their neighbors, even though other traders retained their original rules (Fig. 1). The changed response remained available whenever the relevant neighbor configuration recurred, rather than requiring selected traders to remain in Sell. Comparing a persistent rule change with a one-time instruction to sell separated this continuing influence from the immediate change in current decisions (Supplementary Table S5). The alternative rule screen further tested which responses could produce a majority, so the selected rule was examined within a defined set of local changes rather than treated as a universal description of investor behavior (Supplementary Table S6). The paired design made the no-switch path the comparison for the intervention at each starting market [30, 31]. Repeating the pair across starting markets then measured how frequently a majority formed within the observation window. This interpretation concerned assigned decision rules and their propagation through neighbors; traders did not learn the new rule, choose it for its profitability, or pass the rule itself to other traders.

The decision instability could develop without a price response entering later decisions. Holding the technical traders removed that route of influence, yet the rule change still produced Sell majorities (Fig. 2). In the spatial examples, growing groups of sellers linked local responses to changes across the grid (Fig. 4). Such patterns resemble the growth of ordered regions in other interacting systems away from equilibrium [32, 33], while dependence on the changed fraction connects the model to accounts of how local changes spread through a population [34–37]. The finite-size comparisons described a sharper majority response on larger grids, but the estimated midpoint continued to depend on the observation period (Fig. 3; Supplementary Table S9). A market remaining mixed at one endpoint could therefore require further observation before its later outcome was known. Neither the selected images nor the sampled crossing curves established a threshold for indefinitely large or long-lived markets [27, 28]. The resemblance to droplets forming and joining also did not establish classical nucleation. We measured no energy barrier, surface tension, formation rate, or critical group size separating growth from decay. The supported mechanism was the propagation of changed responses through neighboring decisions over finite grids and observation windows.

The corrected execution mechanism changed the financial interpretation of limited dealer resources. Dealer models commonly connect trading to intermediary cash and inventory [38–41], but holdings alone do not specify how those resources translate into prices. Here, the constant-product reserve curve determined both the uniform price that supported a batch and the marginal quote left after settlement [23]. When the dealer bought shares, its share reserve increased and its cash reserve decreased, so subsequent net sales could execute at lower prices. Shallow positive reserves thus allowed large proportional declines without requiring a large fraction of eligible orders to be rejected (Fig. 2C,D). Zero depth was a separate fixed-price control, because the dealer took no net position and

only opposing trader orders matched. This distinction prevented the zero-depth case from being read as evidence that every small dealer protected the market from a crash. The no-switch comparisons also mattered because shallow markets had large drawdowns while retaining the original decision rule (Supplementary Table S2). The paired increases separated that baseline financial variation from the intervention. The evidence therefore linked crash magnitude to the execution and price mechanism as well as to available resources; resource scarcity alone did not imply a small proportional decline.

Financial conditions could also feed back into the spread of selling. Changing dealer depth or trader holdings altered completed orders and prices, which then changed technical-trader decisions and the local inputs seen by their neighbors (Figs. 2 and 6). Holding those traders prevented the same financial changes from entering the interior update, providing an exact control on this route of influence. With technical traders active, majority probability did not consistently increase or decrease with depth, even though drawdown at the comparison with $f = 0.20$ decreased across positive depths. Freezing the Hold-band width tested one contributor to feedback while preserving responses to a moving reference price (Supplementary Table S7). That comparison did not remove technical trading, and its frozen widths retained differences accumulated during each starting market's baseline. The uncertainty in the paired contrast also prevented a clear attribution of the depth difference to width adaptation. The additional reserve-depth series changed funding and initial price sensitivity together, so it could not identify their independent effects (Supplementary Table S8). This coupling distinguishes the experiment from models that assign a separate impact coefficient to completed order imbalance [42–44]. Majority probability and drawdown consequently measured different outcomes: the share of markets in which enough traders decided to sell, and the size of the largest decline along the resulting price path. Neither outcome alone specified the other.

The model differed from established agent-based markets in where traders interacted and in what was changed. In the minority game, every trader reads the same record of past outcomes, keeps a score for each of its strategies, and uses the one that would have done best, so traders interact only through the market outcome and the main control parameter is the number of distinguishable histories relative to the number of traders [45, 46]. In the Santa Fe artificial stock market, traders learn forecasting rules from a shared price and dividend [5]; adding technical rules to a trader's repertoire was a dominant strategy that lowered wealth and raised volatility for all [47], and the rate at which traders revised their rules settled at an equilibrium that left earnings below what slower revision would have produced [48]. Models with chartists and fundamentalists let traders move between the two groups at rates set by recent profits or by the opinions of others [7, 9, 49]. Lattice and cellular-automaton markets encoded local interactions through stochastic imitation or spin-like responses, sometimes alongside market-wide signals [10, 11, 17–20, 50]. Wolfram's one-dimensional market and a later model built on Rule 110 used deterministic rules and set the price from the numbers of buyers and sellers [14, 15]. These models explained aggregate market behavior through different forms of learning and interaction. The present comparison focused instead on a persistent change assigned to a specified subset of traders. Here every trader kept a fixed rule, the intervention changed one entry of the rule table for a chosen fraction of traders, and the outcome was compared with a market that shared the same history. Only the technical traders on the edge decided from the price, so the same rule change could be tested with those traders active and held. Orders needed cash or shares and were completed by other traders or by a dealer with limited holdings, so a Sell majority and a price crash were measured separately. These choices gave up the learning and strategic behavior of established models in exchange for a rule space that can be enumerated and an intervention that can be repeated exactly. The questions answered were correspondingly narrower: not whether technical trading emerges or whether the market becomes efficient, but how many traders had to change a rule before a Sell majority formed, and whether its formation required feedback from price.

Several modeling choices limit how far these conclusions extend. Traders followed assigned rules rather than choosing strategies to maximize returns, and updated together on a regular grid under one baseline lookup table. Different neighbor configurations could have the same sum and therefore generate the same response. The technical traders followed one band rule and occupied selected perimeter sites, leaving other placements and price-based strategies untested. Unit orders cleared at a common price against a constant-product dealer, a specific reserve mechanism rather than a calibrated representation of an equity exchange. The model had no order book, transaction fees, dividends, borrowing, or strategic dealer risk management, and its time steps and price response were not fitted to market data. Its simulated crashes therefore did not predict the timing or magnitude of actual market crashes. Conservation and the isolated-round-trip identity constrained settlement, but did not

establish economic realism or exclude profits earned from other traders. The long no-switch controls extended observation of existing starting markets without establishing their behavior indefinitely (Appendix A.5). Likewise, neither the baseline period nor an average over final states demonstrated that the distribution of market states had stopped changing. Other decision rules, asynchronous updates, alternative interaction networks, and different execution mechanisms would test how the connection between local decisions and price declines depended on these assumptions.

5 Conclusion

A persistent change in selected traders' local decision rules could produce a Sell majority without feedback from prices. The fraction needed depended on the number of interacting traders and the observation window, while active technical traders allowed financial conditions to modify later decisions. A Sell majority did not determine the price decline by itself: trader cash and shares constrained desired orders, and the dealer's reserve curve determined the prices at which net trades could complete. Under the corrected constant-product mechanism, shallow positive reserves accompanied severe declines, while zero depth permitted only matched trades at a fixed price. Limited resources therefore did not supply a general protection against a simulated crash. The financial consequence of widespread selling depended on how those resources entered execution and price formation.

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Tables

Table S1: Design of the four main campaigns. Active technical traders responded to the marginal quote; held technical traders stayed in Hold. All dealers had finite reserves. The size campaign used active technical traders at $h = 1$ and 100, and held technical traders at $h = 100$. Counts include newly simulated no-switch paths, with one such path shared across fractions within a starting market and condition where the design permits.

Campaign	Grid sizes	Dealer h	Technical traders	Seeds	Simulations
Rule change	64	100	active	50	400
Dealer capacity	64	0–100	active/held	50	9,600
Grid size	32–128	$1, 100$	active/held	50	9,750
Cash and shares	64	$0.1, 1, 100$	active/held	50	10,800

Table S2: Paired rule changes on a 64×64 grid over 3,000 post-switch steps. In A, majority probability p counts paths reaching an interior Sell fraction of at least one-half; median onset is conditional on reaching that value. Undefined means no crossing. In B, DD is median maximum drawdown in percent. ΔDD is the median paired difference, in percentage points, with a pointwise 95% whole-market bootstrap confidence interval (CI). These contrasts separate the rule switch from baseline financial variation.

A. Rule switch at $h = 100$, active technical traders

Fraction f	Seeds	No-switch p	Switched p	Median onset
0.10	50	0.00	0.00	Undefined
0.15	50	0.00	0.00	Undefined
0.20	50	0.00	1.00	1070.0
0.25	50	0.00	1.00	311.5

B. Dealer depth at $f = 0.20$, active technical traders

h	Seeds	No-switch DD	Switched DD	Paired ΔDD [95% CI]
0	50	0.00	0.00	0.00 [0.00, 0.00]
0.1	50	77.04	99.99	22.96 [22.56, 23.74]
0.5	50	42.83	99.82	57.03 [56.25, 57.86]
1	50	32.92	99.52	66.55 [65.56, 67.71]
2	50	23.15	98.59	75.29 [74.78, 75.89]
5	50	14.03	94.43	80.31 [79.22, 80.96]
10	50	8.31	85.68	77.33 [76.71, 77.48]
100	50	0.75	25.15	24.39 [24.20, 24.47]

Table S3: Grid-size comparisons over 50 starting markets and 3,000 post-switch steps. Brackets give 95% confidence intervals (CI) from 2,000 whole-market bootstrap samples. n_{int} counts fractions with $0.05 < \hat{p} < 0.95$. Crossings use first upward linear interpolation. Susceptibility maxima are selected over the sampled grid, including reselection in each bootstrap sample. Undefined denotes an unbracketed crossing; intervals use available bracketed samples.

Technical traders	h	L	n_{int}	f_{50} [95% CI]	Width [95% CI]	Peak χ_L [95% CI]
Active	1	32	9	0.212 [0.208, 0.217]	0.020 [0.014, 0.028]	26.3 [22.2, 34.4]
Active	1	48	6	0.186 [0.184, 0.188]	0.011 [0.008, 0.019]	54.6 [41.3, 69.9]
Active	1	64	9	0.177 [0.175, 0.180]	0.008 [0.007, 0.013]	85.9 [79.9, 114.6]
Active	1	96	5	0.168 [0.167, 0.169]	0.005 [0.004, 0.007]	143.2 [127.4, 202.5]
Active	1	128	7	0.167 [0.165, 0.168]	0.007 [0.005, 0.009]	219.8 [198.5, 300.9]
Active	100	32	6	0.214 [0.210, 0.218]	0.020 [0.012, 0.028]	25.7 [22.2, 33.9]
Active	100	48	5	0.186 [0.184, 0.188]	0.011 [0.008, 0.022]	51.2 [38.6, 65.3]
Active	100	64	5	0.175 [0.174, 0.180]	0.011 [0.009, 0.014]	88.5 [72.2, 124.2]
Active	100	96	5	0.169 [0.165, 0.171]	0.009 [0.007, 0.011]	132.5 [119.7, 178.7]
Active	100	128	6	0.166 [0.165, 0.168]	0.007 [0.005, 0.009]	230.0 [199.8, 302.8]
Held	100	32	5	0.216 [0.211, 0.224]	0.023 [0.018, 0.027]	21.2 [16.5, 27.2]
Held	100	48	5	0.193 [0.184, 0.195]	0.019 [0.016, 0.022]	48.6 [40.3, 65.3]
Held	100	64	5	0.178 [0.176, 0.180]	0.015 [0.008, 0.019]	110.5 [78.5, 138.5]
Held	100	96	5	0.168 [0.167, 0.170]	0.009 [0.006, 0.012]	168.7 [137.6, 221.0]
Held	100	128	7	0.166 [0.165, 0.167]	0.008 [0.004, 0.010]	241.2 [202.6, 326.0]

Table S4: Probability of a Sell majority after changes in trader cash and shares over 50 starting markets and 3,000 post-switch steps. Each fraction includes 27 combinations of cash, shares, and dealer depth. Held decisions gave the same probability across those combinations; active decisions gave the stated range.

Fraction f	Technical traders held	Technical traders active (range)
0.170	0.22	0.12–0.36
0.175	0.34	0.30–0.58
0.180	0.64	0.54–0.80
0.185	0.70	0.72–0.94

Table S5: A one-time instruction to sell versus a persistent rule change at the same targeted traders. Both conditions used $h = 100$, 3,000 post-intervention steps, and matched no-switch controls. Entries are majority probabilities with 95% Wilson intervals.

Technical traders	f	Seeds	Sell pulse	Persistent rule
Active	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Active	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Active	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.15	50	0.00 [0.00, 0.07]	0.00 [0.00, 0.07]
Held	0.20	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]
Held	0.25	50	0.00 [0.00, 0.07]	1.00 [0.93, 1.00]

Table S6: All single-response changes to Rule 19280 with $h = 100$. Each entry gives majority probability with technical traders held / active over 20 starting markets and 3,000 post-switch steps. Outputs are Buy $= 0$, Hold $= 1$, and Sell $= 2$. This descriptive screen used no multiplicity adjustment.

Sum c	Output change	Rule	$f = 0.15$	$f = 0.20$	$f = 0.25$
0	$2 \to 0$	19278	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
0	$2 \to 1$	19279	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 1$	19283	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
1	$0 \to 2$	19286	0.00 / 0.00	1.00 / 1.00	1.00 / 1.00
2	$0 \to 1$	19289	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
2	$0 \to 2$	19298	0.95 / 0.95	1.00 / 1.00	1.00 / 1.00
3	$0 \to 1$	19307	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
3	$0 \to 2$	19334	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
4	$1 \to 0$	19199	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
4	$1 \to 2$	19361	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
5	$1 \to 0$	19037	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
5	$1 \to 2$	19523	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00
6	$2 \to 0$	17822	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
6	$2 \to 1$	18551	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 0$	14906	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
7	$2 \to 1$	17093	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 0$	6158	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00
8	$2 \to 1$	12719	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00

Table S7: Adjusting versus fixed Hold-band widths at $f = 0.175$ over 50 starting markets. Brackets in the rows give 95% Wilson confidence intervals (CI). Sell is the mean fraction of technical traders in Sell over post-switch steps and starting markets. The moving center continued to update. The $h = 0.1$ minus $h = 100$ probability difference was -0.06 with adaptive widths and -0.26 with frozen widths. Their paired difference was 0.20, with a 95% interval [-0.08, 0.50] from 2,000 whole-market bootstrap samples.

h	Adaptive p [95% CI]	Frozen p [95% CI]	Adaptive Sell	Frozen Sell
0	0.34 [0.22, 0.48]	0.34 [0.22, 0.48]	0.000	0.000
0.1	0.44 [0.31, 0.58]	0.22 [0.13, 0.35]	0.139	0.044
0.5	0.36 [0.24, 0.50]	0.38 [0.26, 0.52]	0.123	0.081
1	0.36 [0.24, 0.50]	0.40 [0.28, 0.54]	0.128	0.086
2	0.44 [0.31, 0.58]	0.40 [0.28, 0.54]	0.117	0.094
5	0.38 [0.26, 0.52]	0.36 [0.24, 0.50]	0.116	0.140
10	0.40 [0.28, 0.54]	0.38 [0.26, 0.52]	0.125	0.163
100	0.50 [0.37, 0.63]	0.48 [0.35, 0.61]	0.103	0.168

Table S8: Additional reserve-depth comparisons at $f = 0.20$ with active technical traders and 50 starting markets over 3,000 post-switch steps. The initial log-price slope per normalized net purchase is approximately $2/h$; this is implied by the reserve curve, not a separate price-impact coefficient. Depth changes funding and price sensitivity together. Drawdown is a percentage, and rejection uses post-switch eligible orders. Probability brackets are 95% Wilson intervals.

h	Initial slope $2/h$	Majority probability	Median drawdown (%)	Rejection
40	0.050	1.00 [0.93, 1.00]	48.62	1.41×10^{-6}
100	0.020	1.00 [0.93, 1.00]	25.15	6.04×10^{-7}
400	0.005	1.00 [0.93, 1.00]	7.08	2.82×10^{-7}

Table S9: Observation time and the required Sell fraction with technical traders held and $h = 100$. Entries are interpolated f_{50} with 95% whole-market bootstrap intervals over 50 starting markets. The fraction grid is independent of adaptive refinement. Undefined denotes an unbracketed crossing; intervals use available bracketed samples.

L	Post-switch steps	Criterion 0.4	Criterion 0.5	Criterion 0.6
32	750	0.236 [0.228, 0.238]	0.249 [0.245, 0.255]	0.266 [0.260, 0.270]
32	1,500	0.216 [0.210, 0.220]	0.230 [0.228, 0.234]	0.245 [0.238, 0.250]
32	3,000	0.207 [0.205, 0.210]	0.219 [0.210, 0.225]	0.229 [0.226, 0.232]
64	3,000	0.172 [0.169, 0.176]	0.178 [0.176, 0.180]	0.186 [0.180, 0.188]
64	9,000	0.165 [0.163, 0.167]	0.167 [0.165, 0.168]	0.171 [0.169, 0.175]
128	3,000	0.163 [0.162, 0.164]	0.167 [0.165, 0.168]	0.169 [0.168, 0.171]
128	6,000	0.154 [0.153, 0.156]	0.157 [0.156, 0.158]	0.159 [0.158, 0.160]
128	12,000	0.150 [0.148, 0.151]	0.152 [0.150, 0.153]	0.153 [0.152, 0.155]

Table S10: Logistic fits and sampled susceptibility maxima over 50 starting markets and 3,000 post-switch steps. Parentheses give whole-market bootstrap standard errors (SE). The fitted curve imposes increasing probability; empirical crossings do not. The final column gives a sampled fraction, whose precision is limited by the local fraction grid.

Technical traders	h	L	Logistic f_{50} (SE)	Logistic width (SE)	f at peak
Active	1	32	0.212 (0.0015)	0.020 (0.0017)	0.217
Active	1	48	0.185 (0.0013)	0.014 (0.0012)	0.190
Active	1	64	0.177 (0.0006)	0.010 (0.0007)	0.185
Active	1	96	0.168 (0.0005)	0.006 (0.0006)	0.175
Active	1	128	0.166 (0.0005)	0.006 (0.0005)	0.168
Active	100	32	0.215 (0.0015)	0.018 (0.0017)	0.217
Active	100	48	0.185 (0.0011)	0.013 (0.0014)	0.190
Active	100	64	0.177 (0.0008)	0.009 (0.0007)	0.190
Active	100	96	0.168 (0.0006)	0.007 (0.0005)	0.175
Active	100	128	0.166 (0.0004)	0.005 (0.0004)	0.168
Held	100	32	0.218 (0.0016)	0.018 (0.0014)	0.225
Held	100	48	0.189 (0.0011)	0.013 (0.0009)	0.190
Held	100	64	0.178 (0.0010)	0.011 (0.0008)	0.185
Held	100	96	0.169 (0.0006)	0.007 (0.0006)	0.172
Held	100	128	0.166 (0.0005)	0.006 (0.0004)	0.168

Table S11: Longer observation with $h = 100$ and active technical traders. Times are steps after the rule change, following 1,000 baseline steps. Drawdowns and price changes are percentages; price changes use the quote at activation as reference. Final Sell averages the last 500 steps of the longer window. None means no Sell majority by 10,000 steps. Selection used newly computed short-run summaries before inspection of images. The extensions follow the same starting markets, and are not independent replications.

A. Five examples selected from new 3,000-step results

f	Seed	First majority	Drawdown 3,000 steps	Drawdown 10,000 steps	Price change 10,000 steps	Final Sell
0.150	2	None	5.22	5.22	-3.29	0.268
0.175	15	4,073	10.17	26.34	-26.32	0.897
0.175	41	2,330	22.14	25.51	-25.49	0.594
0.185	5	1,836	23.77	26.07	-26.03	0.910
0.200	38	1,028	25.61	26.50	-26.48	0.913

B. All 50 starting markets at $f = 0.150$

Decision rule	Steps	Majorities / 50	Median drawdown	Worst drawdown	Median price change
New rule at 15%	3,000	0/50	5.04	12.29	-3.40
New rule at 15%	5,000	0/50	5.25	17.65	-3.49
New rule at 15%	10,000	2/50	5.87	25.30	-3.54
Original rule	3,000	0/50	0.75	1.15	0.06
Original rule	5,000	0/50	0.81	1.15	0.04
Original rule	10,000	0/50	0.88	1.15	0.04

Figures

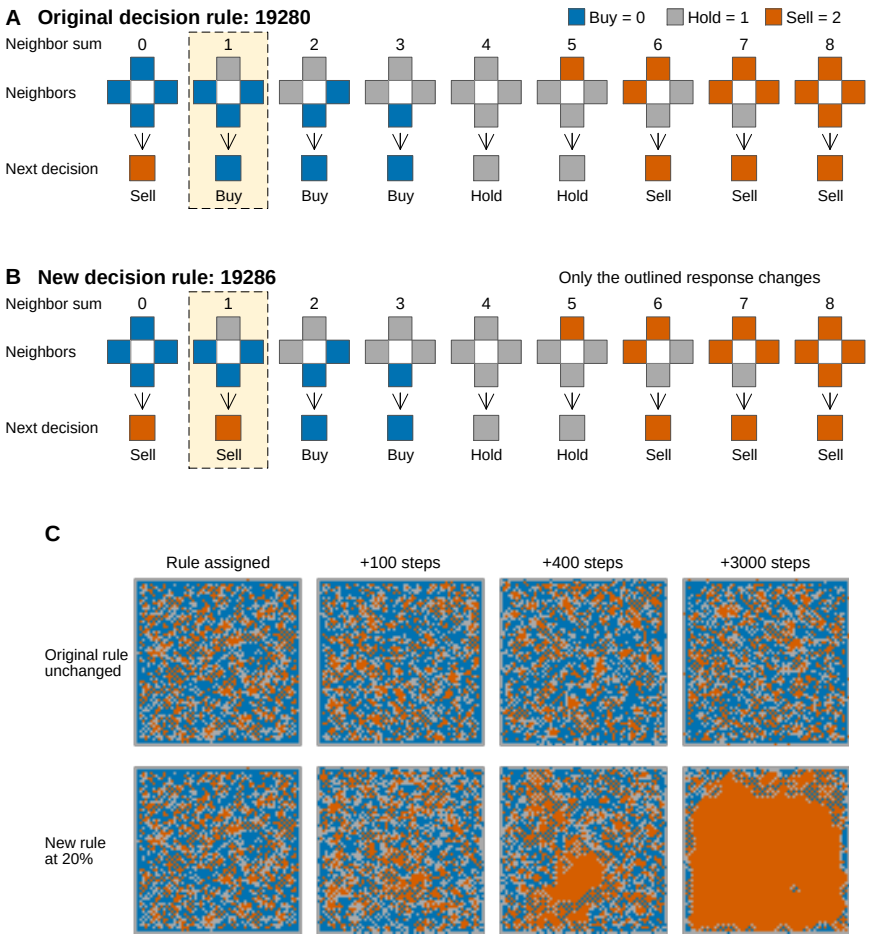


Figure 1: The changed response and paired market decisions. **(A)** Baseline Rule 19280. **(B)** Activated Rule 19286. Each cross contains the four neighbors used to determine the next decision, shown below its arrow. The hollow center is the trader being updated; its current decision does not enter the lookup table. Blue, gray, and orange denote Buy (0), Hold (1), and Sell (2). One arrangement represents each neighbor sum; every arrangement with that sum produces the same response. The outlined column is the only rule change: three Buys and one Hold produce Buy under the baseline and Sell under activation. **(C)** Paired 64×64 grids at activation and after 100, 400, and 3,000 steps. The upper row retains the baseline rule; the lower assigns the new rule to 20% of interior traders. Both paths share a 1,000-step baseline, $h = 100$, and active technical traders. Starting market 14 was selected from 50 by proximity to the median paired difference in peak Sell fraction. Ensemble outcomes and conditional onset times appear in Supplementary Table S2.

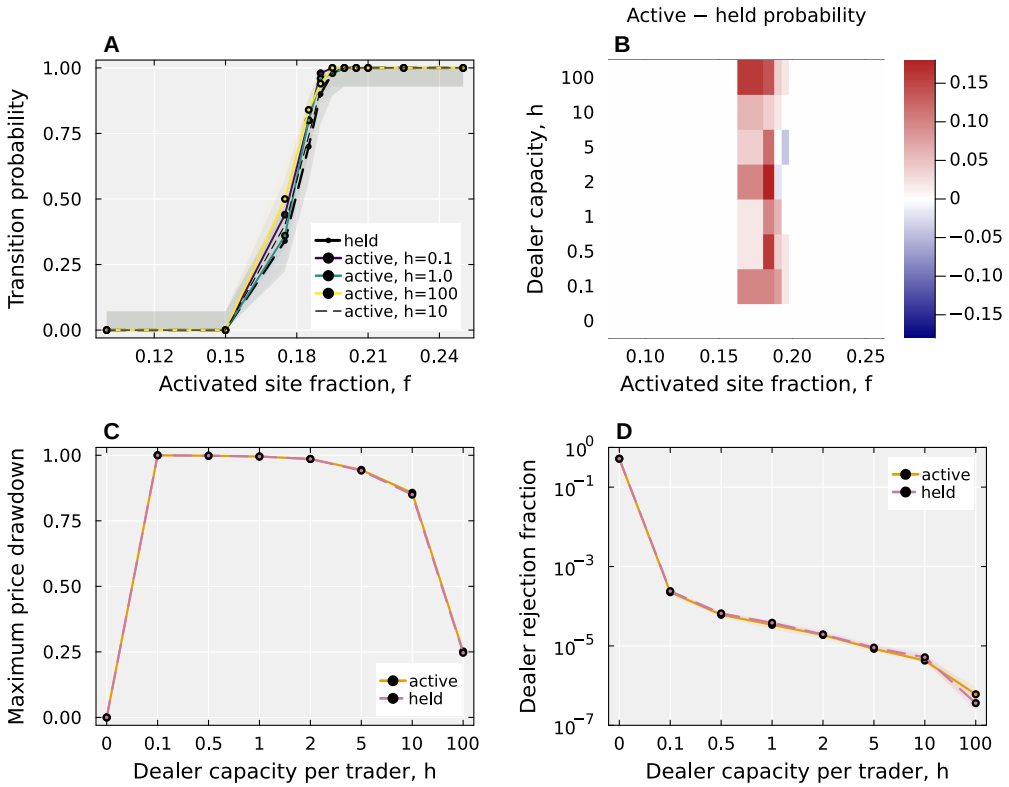


Figure 2: Rule changes across dealer reserve depths over 3,000 post-switch steps. **(A)** Probability of an interior Sell majority with technical traders held or active. Bands are 95% Wilson intervals over 50 starting markets. **(B)** Active-minus-held majority probability at each depth and target fraction. **(C)** Median maximum drawdown at $f = 0.20$, with interquartile bands. **(D)** Median fraction of post-switch eligible orders rejected at $f = 0.20$, on a logarithmic axis with interquartile bands. Depth h is initial dealer shares per trader and also determines price sensitivity through the reserve curve. The $h = 10$ dashed overlay in A is a distinct finite-depth comparison; the deep arm is $h = 100$. At zero depth only opposing trader orders match and price stays fixed. Panels C and D describe switched markets. Matched no-switch drawdowns and paired differences appear in Supplementary Table S2.

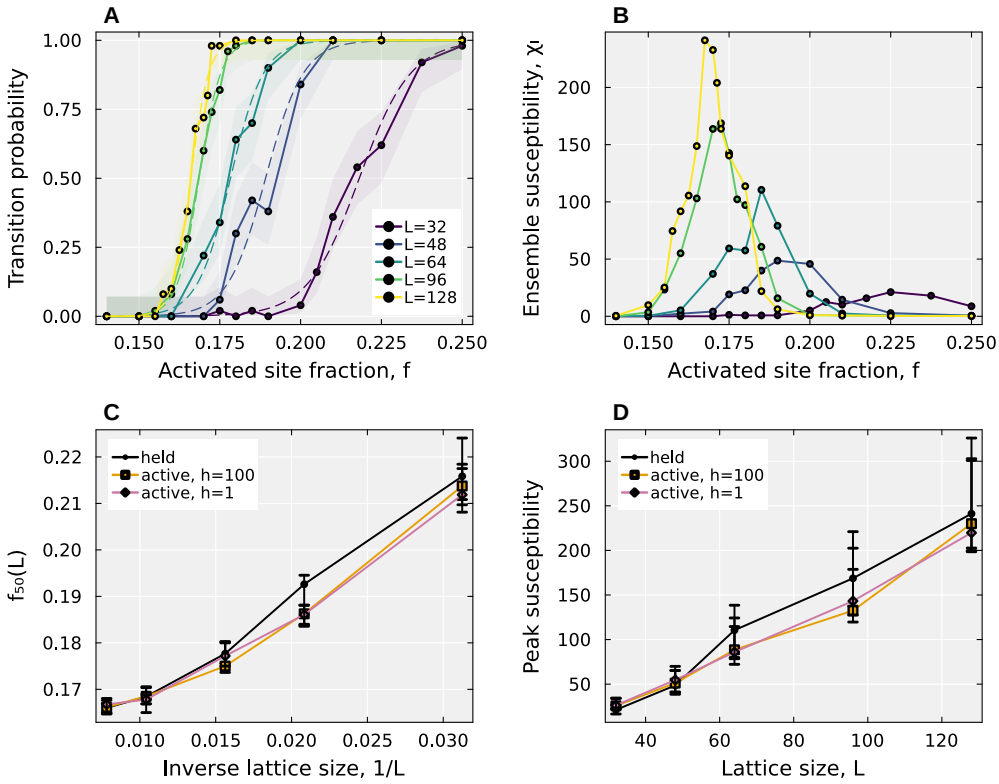


Figure 3: Market size and the formation of a Sell majority over 50 starting markets and 3,000 post-switch steps. **(A)** Majority probability with technical traders held, with 95% Wilson bands. **(B)** Susceptibility with technical traders held: the variance across markets of the final 500-step mean Sell fraction, multiplied by the number of interior traders. **(C)** Interpolated fraction $f_{50}(L)$ at which half the markets reached a majority, against inverse grid size, for held traders at $h = 100$ and active traders at $h = 1$ and $h = 100$. **(D)** Sampled maximum susceptibility versus grid size. Error bars in C and D are 95% whole-market bootstrap intervals. Solid lines connect tested values; dashed curves in A are logistic fits. No law for indefinitely large grids was fitted.

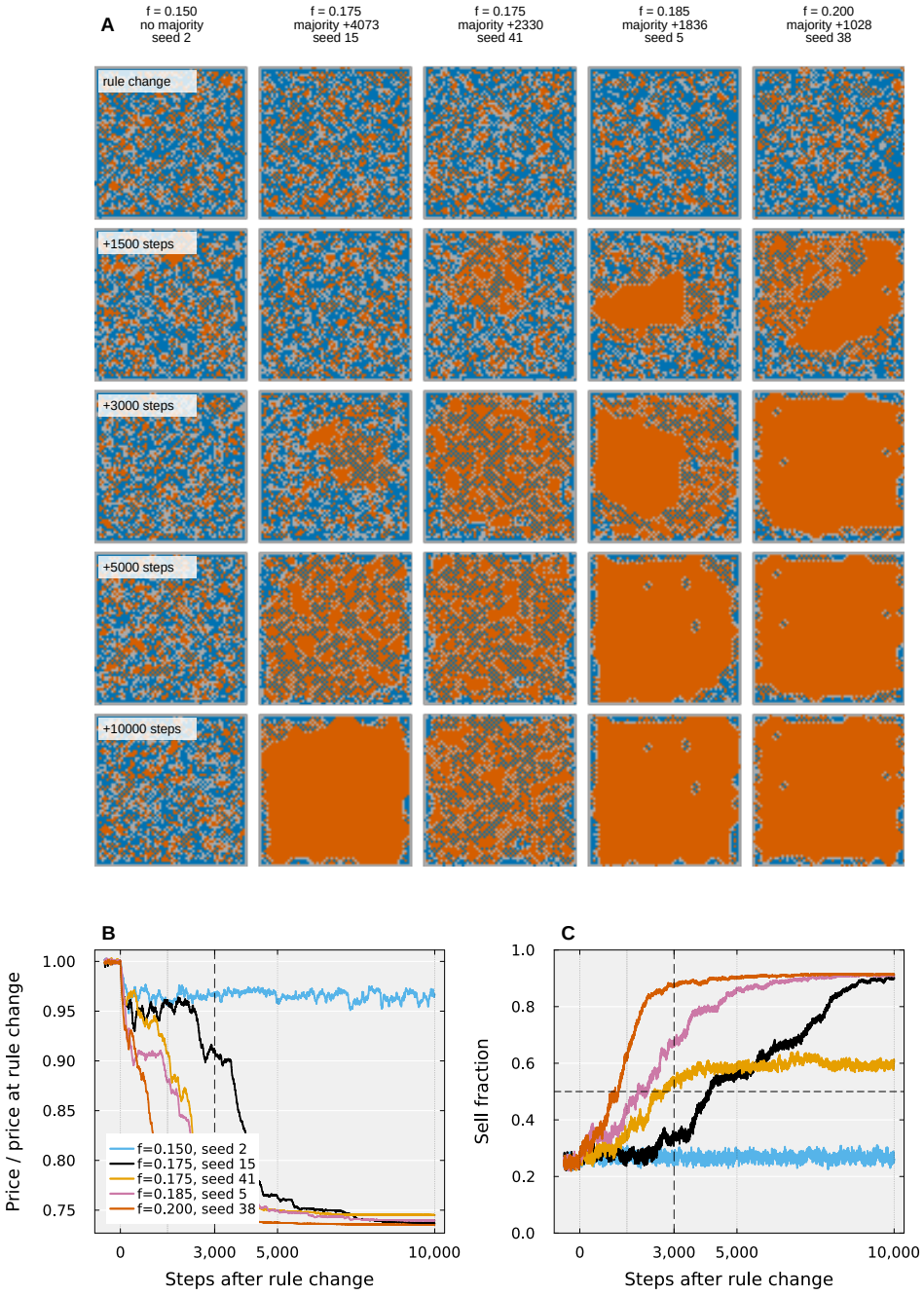


Figure 4: Longer histories of five numerically selected markets with active technical traders and $h = 100$. Blue, gray, and orange denote Buy, Hold, and Sell. **(A)** Grid decisions at activation and after 1,500, 3,000, 5,000, and 10,000 steps. Selection used the newly computed 3,000-step summaries before inspection of images. At that endpoint, the first two markets had not reached a majority and the other three had. Nonmajority examples were nearest their group's median final Sell fraction; majority examples were nearest their group's median onset, with seed breaking ties. Starting markets were 2, 15, 41, 5, and 38, from left to right. The longer outcomes did not affect selection. **(B)** Marginal quote divided by its value at activation. **(C)** Interior Sell fraction, with one-half marked by a horizontal line. The dark vertical dashed line marks the original 3,000-step endpoint; other guides mark the remaining snapshots. Onsets and financial summaries appear in Supplementary Table S11.

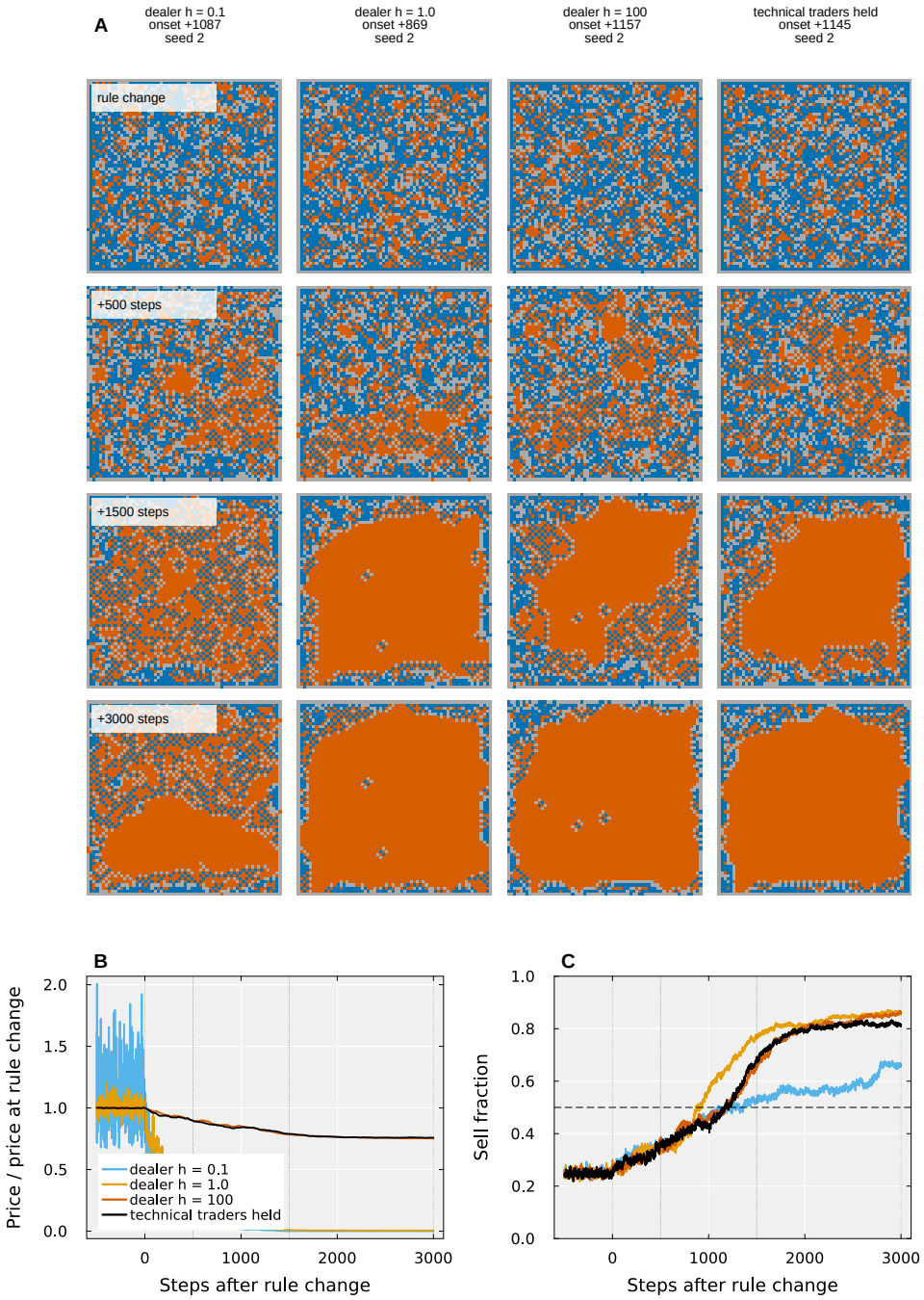


Figure 5: The same starting market at different dealer depths with $f = 0.20$. **(A)** Grid decisions at activation and after 500, 1,500, and 3,000 steps. The first three columns have active technical traders at $h = 0.1$, 1, and 100; the last holds the technical traders at $h = 100$. Starting market 2 was selected among markets reaching a majority at all three active depths by minimizing the summed distance from their depth-specific median onsets, with seed breaking ties. **(B)** Marginal quote relative to activation. **(C)** Interior Sell fraction. The held condition prevents price feedback into decisions while retaining full trading and price calculations. The selected paths illustrate the same initialization across conditions; ensemble probabilities and drawdowns appear in Fig. 2.

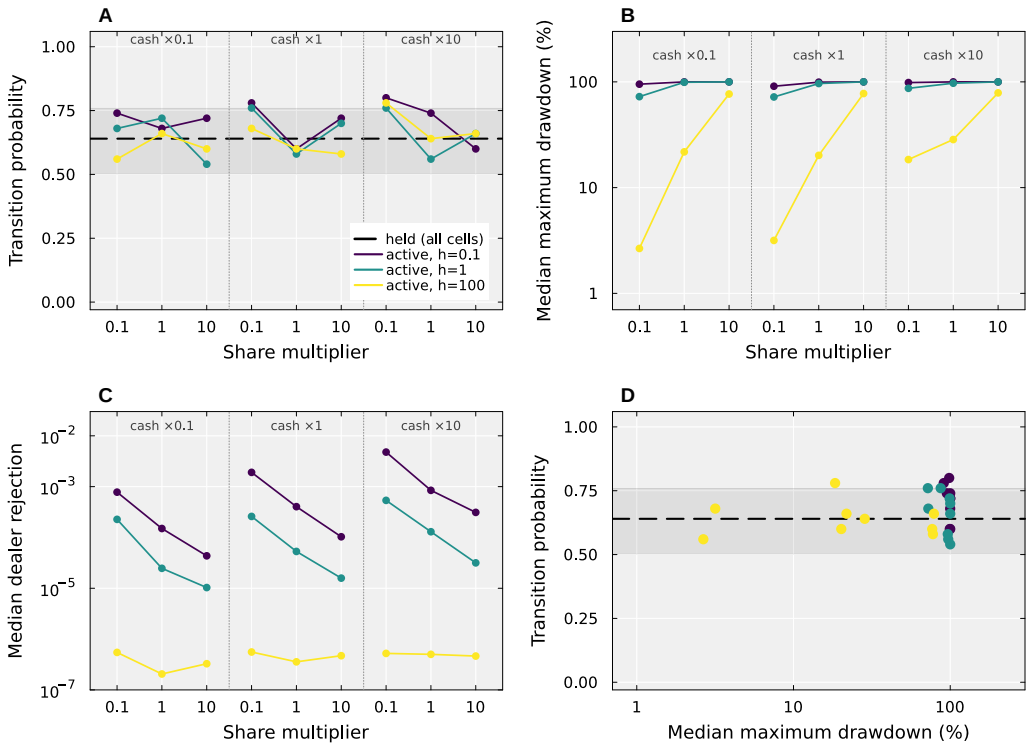


Figure 6: Changing trader cash and shares at the rule switch, with active technical traders, $f = 0.18$, and 50 starting markets over 3,000 post-switch steps. In A–C, the nine resource combinations are grouped by cash multiplier, with share multiplier on the axis and a series for each dealer depth. **(A)** Majority probability. The dashed line and band give the held-condition probability and its 95% Wilson interval, identical across financial conditions. **(B)** Median maximum drawdown on a logarithmic axis. **(C)** Median fraction of post-switch eligible orders rejected, on a logarithmic axis. **(D)** Majority probability against median drawdown for all 27 active combinations, with the held reference repeated. The drawdown axis follows B. Probability axes cover the full possible range; all displayed drawdowns and rejection medians are strictly positive.

A Supplementary methods and validation

A.1 One complete simulation step

The simulation separated a desired decision from its execution. Orders used the decisions stored before the interior update, while technical traders used the new marginal quote before the next interior decisions were calculated. A resource shortage could therefore stop an order without changing the state seen by neighboring traders. The uniform execution price depended on the net quantity supported by the entire batch (Eqs. 3 and 4), and differed from the marginal quote left after settlement (Eq. 7). Executed counts specified the net transfer and its normalized imbalance (Eqs. 5 and 8), which changed the dealer reserves (Eq. 6). The new quote then updated the reference band (Eq. 9). Algorithms S1 and S2 specify this ordering for the main persistent-rule experiments with adaptive bands and no cost-basis restriction. The pulse, alternative-rule, and frozen-band controls modify the corresponding updates as described in Appendix A.5.

Algorithm S1 One market step, $t-1 \to t$, in simulation order. Stored decisions generated orders. Clearing determined a common execution price, and the settled dealer reserves set the new marginal quote.

Require: grid X_{t-1}; technical-trader decisions; trader books; dealer (W_{t-1}, Q_{t-1}); quote and band state $(P_{t-1}, \mu_{t-1}, \sigma_{t-1})$; rule field R_i; parameters; seed
Ensure: grid X_t; technical-trader decisions; settled books and dealer; quote and band state (P_t, μ_t, σ_t); order counts
if t is the scheduled rule-switch step **then**
  $R_i \leftarrow 19286$ at the preselected target sites {preserve current decisions and financial state}
end if
if t is a scheduled resource intervention **then**
  for all traders i **do**
    $w_i \leftarrow g_w w_i$; $q_i \leftarrow \mathrm{round}(g_q q_i)$
  end for
end if
call Algorithm S2 using the stored decisions
receive settled books, dealer, order counts, execution price F_t, and quote P_t
$\mu_t \leftarrow \mu_{t-1} + \alpha(P_t - \mu_{t-1})$
$\sigma_t \leftarrow \sigma_{t-1} + \alpha(|P_t - \mu_t| - \sigma_{t-1})$
if technical traders are active **then**
  for all technical traders i **do**
    $e_i(t) \leftarrow 0$ if $P_t < \mu_t - k_i\sigma_t$; 2 if $P_t > \mu_t + k_i\sigma_t$; 1 otherwise
  end for
else
  $e_i(t) \leftarrow 1$ for every technical trader i
end if
write $e_i(t)$ to the selected perimeter sites; leave other perimeter sites in Hold
for all interior traders i, simultaneously **do**
  $c_i \leftarrow \sum_{j \in \partial i} s_j$ using old interior and new technical-trader decisions
  $s_i(t) \leftarrow \mathcal{R}_{R_i}(c_i)$
end for

Algorithm S2 Uniform-price reserve clearing without a cost-basis restriction. The binary search found the greatest supported integer net purchase in the direction of initial excess demand.

Require: desired unit buys and sells from stored decisions; trader books $(w_i, q_i, \bar{c}_i)$; dealer (W, Q); previous quote; seed and step
Ensure: settled books and dealer; order counts; execution price and next quote
if $h = 0$ **then**
  use the previous quote; match opposing affordable buys and supplied sells
  apply seeded priorities if needed; settle matched orders; keep the quote fixed
  return settled books and order counts
end if
define $F(x) = W/(Q - x)$ for integer x satisfying $Q - x \geq 1$
define $B(x)$ as desired buyers with $w_i \geq F(x)$ and $S(x)$ as desired sellers with $q_i \geq 1$
$d \leftarrow \operatorname{sign}[B(0) - S(0)]$; $\ell \leftarrow 0$
$u \leftarrow \min(B(0) - S(0), Q - 1)$ if $d > 0$, otherwise $S(0) - B(0)$
while $\ell < u$ **do**
  $k \leftarrow \lfloor (\ell + u + 1)/2 \rfloor$
  if $d[B(dk) - S(dk)] \geq k$ **then**
    $\ell \leftarrow k$
  else
    $u \leftarrow k - 1$
  end if
end while
$x \leftarrow d\ell$; $F_t \leftarrow F(x)$
collect eligible sets $\mathcal{B}$ and $\mathcal{S}$ at F_t
$n^B \leftarrow \min(|\mathcal{B}|, |\mathcal{S}| + x)$; $n^S \leftarrow n^B - x$
suppression $\leftarrow$ desired counts minus eligible counts
rejection $\leftarrow$ eligible counts minus executed counts
rank any rationed side by $z(\text{seed}, t, i, \text{side})$, breaking ties by identity
for all selected buyers i **do**
  $\bar{c}_i \leftarrow (\bar{c}_i q_i + F_t)/(q_i + 1)$; $q_i \leftarrow q_i + 1$; $w_i \leftarrow w_i - F_t$
end for
for all selected sellers i **do**
  $q_i \leftarrow q_i - 1$; $w_i \leftarrow w_i + F_t$
end for
$Q' \leftarrow Q - x$; $W' \leftarrow WQ/Q'$; $P_t \leftarrow W'/Q'$
return settled books, dealer, counts, F_t, and P_t

A.2 Rationing and paired reproducibility

Each eligible order on a rationed side received a 64-bit score from a SplitMix64 finalizer applied to the simulation seed, step, trader identity, and order side. Sorting by score, then identity, made rationing reproducible while allowing priority to change over time. Paired markets used the same score for the same trader and step even after their eligible sets diverged. Scoring consumed no draws from the generator used for initialization. For paired rule changes, comparisons checked the pre-switch grid and Sell history, technical-trader coordinates and multipliers, and price history through the first post-switch quote. A pure rule switch preserved that first quote because the stored decisions still generated the same orders. A one-time Sell pulse and a resource intervention could change the first batch and therefore were required to agree only before the intervention. For held technical traders, resource comparisons also required identical Sell histories and final grids across all cash and share multipliers.

A.3 Conservation and limiting cases

Let total trader cash and shares be the sums over all traders. Outside scheduled resource interventions, their combination with dealer reserves satisfied the conservation identities:

$$\sum_i w_i(t) + W_t = \sum_i w_i(0) + W_0,$$
$$\sum_i q_i(t) + Q_t = \sum_i q_i(0) + Q_0.$$

The production wrapper checked these identities separately before and after an intentional resource change. Share conservation was exact. The allowed absolute cash discrepancy was the larger of 10^{-5} and 10^{-11} times total cash at the beginning of the checked interval. For positive depth, checks required a finite positive quote, nonnegative trader resources, preservation of the reserve product to relative 10^{-9}, and agreement between the quote and reserve ratio. The implementation evaluated new dealer cash as WQ/Q' to limit accumulated rounding error; the equivalent transfer was $F_t x_t$. These checks ran after every newly completed simulation, before its execution record was written.

Zero depth permitted only matched trader orders and kept price fixed. Technical traders then remained in Hold under either feedback condition, providing an exact comparison between active and held interior decisions. At positive depth, tests checked settlement at the selected price, affordability after clearing, discrete rationing, and isolated round trips that restored dealer inventory and cash. They also covered delayed activation, preservation of a common prehistory, and the held-decision invariance across financial interventions. An independent implementation of the held decision update served as a test oracle. Every production condition, including held decisions, was nevertheless rerun with the full financial simulator. Finite depths were distinct model conditions: no finite-depth path was assumed to equal an unlimited dealer.

A.4 Simulation design and reproducibility

The clean rebuild began with an empty data and figure tree and a frozen copy of the corrected simulator, drivers, parameter manifest, and analysis code. No earlier simulation output or checkpoint was imported. Completed simulations wrote an execution ledger, while phase manifests and source hashes constrained any resumption to outputs created within this rebuild. All retained experiments were rerun, including held conditions whose decisions could be predicted without calculating finances. Seeds were retained to reproduce the planned starting markets; this was a complete recomputation under the corrected model, not an independent sample of starting markets. The corrected model replaced settlement at a standing quote followed by a separate exponential price update. That earlier mechanism admitted profitable isolated round trips, and its financial outputs did not enter this manuscript. Reserve clearing changed both settlement and price formation, so differences from the earlier model cannot be attributed to one parameter alone.

The four main campaigns comprised 30,550 simulations over 50 starting markets per condition and 3,000 post-switch steps (Supplementary Table S1). The selected paired-switch campaign used $L = 64$, $h = 100$, and active technical traders at fractions 0.10, 0.15, 0.20, and 0.25. Dealer comparisons used fractions 0.10, 0.15, 0.175, 0.185, 0.190, 0.195, 0.200, 0.205, 0.210, 0.225, and 0.250 at $h = 0, 0.1, 0.5, 1, 2, 5, 10, 100$ with technical traders active and held. A newly simulated no-switch path was shared across fractions within each depth, feedback condition, and starting market, avoiding duplicates within the same campaign. The size campaign used $L = 32, 48, 64, 96, 128$ at fractions 0.140, 0.150, 0.160, 0.170, 0.175, 0.180, 0.185, 0.190, 0.200, 0.210, 0.225, and 0.250 before refinement. Its conditions were held technical traders at $h = 100$ and active technical traders at $h = 1$ and $h = 100$. The cash and share campaign crossed independent multipliers 0.1, 1, and 10 with fractions 0.170, 0.175, 0.180, and 0.185, depths 0.1, 1, and 100, and both feedback conditions. Dealer resources increased with the trader count; no size series with fixed absolute dealer resources was included.

The supplementary controls retained these starting-market conventions while varying the intervention, reserve depth, band adaptation, or observation window. Their fixed design preceded this clean rebuild; adaptive refinement and example selection followed the numerical rules specified above. We used Julia 1.12.7 with the repository environment manifest, twelve simulation threads, and one garbage-collection thread. A seeded Xoshiro generator supplied setup draws [51]. Checkpointed drivers wrote atomic records and reconstructed a deterministic row order. A recorded scheduling-only

revision assigned supplementary jobs to available workers individually. Validation against serial execution and the original schedule reproduced complete market histories exactly; the simulation equations, parameters, and analysis were unchanged. Python performed the bootstrap analysis and generated tables from the new summaries. The accompanying repository snapshot contains source hashes, parameter manifests, execution records, package versions, and derived outputs. The project repository is `https://github.com/varnerlab/Wolfram-markets-abm-paper`. The repository retains earlier implementations and outputs as a record of model development, outside the rebuild's permitted data inputs.

The example trajectories used newly computed 3,000-step summaries to select typical markets before inspecting their images. We reran the five selected conditions and all 50 starting markets at $f = 0.150$ for 10,000 post-switch steps, each with a matched no-switch path (Fig. 4; Supplementary Table S11). Selection was not repeated using the longer histories. Comparisons at the original endpoint checked reproduction of the new short-run summaries; additional figure reruns checked full histories and final grids for the selected cases. The longer runs extended observations of the same starting markets and did not supply independent replications. Majority onset began with the first updated state after activation, and all reported execution counts and rejection fractions used post-switch steps. Keeping these measurement windows common avoided mixing baseline trading into one campaign's financial summaries but not another's.

The completed rebuild was checked against its fixed parameter manifest and its execution ledger. All fixed conditions were present, and the new held size and horizon campaigns agreed for 1,800 overlapping starting-market and fraction combinations. Across the main dealer, size, and resource campaign summaries, the largest absolute cash discrepancy was 3.40×10^{-6}, and share discrepancies were zero. These numerical checks accompanied the per-simulation reserve and conservation checks described above.

A.5 Additional controls and sensitivity analyses

The supplementary controls used the same initialization and baseline as the main comparisons, with $L = 64$, $h = 100$, and 3,000 post-intervention steps unless otherwise specified. Every treated path had a newly simulated no-switch comparison. Held conditions calculated full financial histories as well as decisions. A Sell pulse wrote Sell at the target sites immediately before the first post-baseline orders, while retaining the baseline rule for later updates. Sites already in Sell were unchanged. A persistent intervention instead changed the selected rule without overwriting decisions, so its first price remained identical to the no-switch price. These paired interventions compared the same target fractions and starting markets under both feedback conditions (Supplementary Table S5). The rule screen activated all eighteen ways of changing one response in the baseline lookup table, using twenty starting markets per condition and three target fractions. Each screen entry was descriptive rather than a separate confirmatory hypothesis test (Supplementary Table S6).

The band control froze the recent-departure scale at its value immediately before activation, while each technical trader retained its multiplier and the reference price continued to update. Adaptive and frozen paths shared their complete pre-switch histories. The frozen widths could therefore differ across capacities and starting markets. At $f = 0.175$, the estimated majority-probability difference between $h = 0.1$ and $h = 100$ was -0.06 with adaptive widths and -0.26 with frozen widths. Subtracting those two differences gave 0.20, with a 95% whole-market bootstrap interval of [-0.08, 0.50] (Supplementary Table S7). This comparison changed width adaptation while retaining the response to departures from a moving reference price. It therefore tested one part of the price feedback rather than removing technical trading.

Additional depth comparisons used $h = 40, 100, 400$ at $f = 0.20$ with active technical traders (Supplementary Table S8). For the reserve curve, differentiating log price with respect to normalized net trader purchases at initialization gives $2N_{\mathrm{tr}}/H$, approximately $2/h$. The chosen depths therefore spanned initial slopes 0.05, 0.02, and 0.005. No independent exponential impact coefficient was varied: changing depth also changed dealer funding. These comparisons replaced an impact-coefficient sweep belonging to the superseded standing-quote model. They measured sensitivity within the corrected mechanism and did not hold available reserves fixed while changing price sensitivity. Neither the quote nor the time step was calibrated to an observed market.

The horizon campaign used fifty starting markets at fractions 0.140, 0.145, through 0.250, extending to 0.300 for $L = 32$ to bracket short-horizon responses. Held technical traders prevented financial feedback throughout. Endpoint comparisons used the original 3,000 steps and size-dependent windows $3000(L/64)^z$ for $z = 1, 2$, alongside longer observations. Majority criteria of 0.4, 0.5, and 0.6 were extracted from each history. Interpolation and bootstrap intervals were calculated separately for every size, horizon, and criterion (Supplementary Table S9). The no-switch extension followed ten starting markets per feedback condition for 100,000 additional steps. The observed outcomes were active technical traders: 0/10 majorities, 95% Wilson interval [0.000, 0.278]; held technical traders: 0/10 majorities, 95% Wilson interval [0.000, 0.278]. These extensions reused starting markets and did not constitute independent replications. The observed crossing frequencies described the stated interval without establishing behavior at indefinitely long times.

We also characterized the shared pre-switch histories at $h = 100$ with active technical traders. Across the fifty starting markets, mean interior Sell fractions were 0.2494 over steps 501–750 and 0.2504 over steps 751–1,000. In the latter interval, the median within-market log-return standard deviation was 0.000323, and median within-market first and ninety-ninth percentiles were -0.000730 and 0.000729. The mean fraction of exactly zero computed log returns was 0.007. These summaries described the starting markets without establishing stationarity or matching a calibrated empirical return distribution. The post-switch no-change comparisons separately measured baseline financial variation across dealer depths (Supplementary Table S2).

The refined finite-size analysis resampled entire starting markets across fractions in 2,000 bootstrap replicates. The empirical estimate used the first upward crossing and allowed local decreases in probability, while the logistic fit imposed a smooth increasing response (Supplementary Table S10). Each replicate reselected the susceptibility maximum over the sampled grid. Consequently, intervals included variation in which sampled fraction supplied that maximum, but not interpolation bias, uncertainty from selecting the refinement grid, or changes from observing for longer. The recorded valid-bootstrap fraction identified samples in which the requested crossings remained bracketed. No susceptibility exponent or infinite-size limit was fitted.